INTRODUCTION TO LaTeX

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October 2024

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Introduction

LaTeX's features for typesetting mathematics make it a compelling choice for writing technical documents. This article shows the most basic commands needed to get started with writing maths using LaTeX.
Writing basic equations in LaTeX is straightforward, for example:

```
\documentclass{article}
\begin{document}
The well known Pythagorean theorem \(x^2 + y^2 =
z^2 n) was proved to be invalid for other exponents.
Meaning the next equation has no integer
solutions:
\[ x^n + y^n = z^n n]
\end{document
```

The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents.
Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n$$

Mathematical modes: Inline math mode

- One of the main advantages of LaTeX is the ease with which mathematical expressions can be written. LaTeX provides two writing modes for typesetting mathematics:
  - *inline* math mode used for writing formulas that are part of a paragraph
  - *display* math mode used to write expressions that are not part of a text or paragraph and are typeset on separate lines

Inline math mode::

You can use any of these "delimiters" to typeset your math in inline mode:

- \(...\)
- $...$
- \begin{math}...\end{math}

They all work and the choice is a matter of taste, so let's see some examples.

```
\documentclass[12pt, a4paper]{article}
\begin{document}
In physics, the mass-energy equivalence is
stated by the equation $E=mc^2$, discovered in
1905 by Albert Einstein.
\end{document
```

In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Mathematical modes: Inline math mode

To typeset inline-mode math you can use one of these delimiter pairs: \(... \), $ ... $ or ..., as demonstrated in the following example:

```
\documentclass[12pt, a4paper]{article}
\begin{document}
\begin{math}
E=mc^2 is typeset in a paragraph using inline
math mode---as is $E = mc^2$, and so too is
$E = mc^2$.
\end{document
```

$E = mc^2$ is typeset in a paragraph using inline math mode—as is $E = mc^2$, and so too is $E = mc^2$.

Mathematical modes: Display math mode

Equations typeset in display mode can be numbered or unnumbered, as in the following example:

```
\documentclass[12pt, a4paper]{article}
\begin{document}
In physics, the mass-energy equivalence is
stated by the equation \[E=mc^2\], discovered
in 1905 by Albert Einstein.
In natural units ($c = 1$), the formula
expresses the identity
\begin{equation}
     E=m
\end{equation}
\end{document
```

In physics, the mass-energy equivalence is stated by the equation

$$E = mc^2$$

discovered in 1905 by Albert Einstein. In natural units ($c = 1$), the formula expresses the identity

$$E = m \tag{1}$$

More complete examples

The following examples demonstrate a range of mathematical content typeset using LaTeX.

```
\documentclass[12pt, a4paper]{article}
\begin{document}
Subscripts in math mode are written as $a_b$ and superscripts are written as $a^b$.
These can be combined and nested to write expressions such as
\[ T^{i_1 i_2 \dots i_p}_{j_1 j_2 \dots j_q} =
T(x^{i_1},\dots,x^{i_p},e_{j_1},\dots,e_{j_q}) \]
We write integrals using $\int$ and fractions using $\frac{a}{b}$.  Limits are
placed on integrals using superscripts and subscripts:
\[ \int_0^1 \frac{dx}{e^x} = \frac{e-1}{e} \[
Lower case Greek letters are written as $\omega$ $\delta$ etc.  while upper case
Greek letters are written as $\Omega$ $\Delta$.
Mathematical operators are prefixed with a backslash as $\sin(\beta)$,
$\cos(\alpha)$, $\log(x)$ etc.
\end{document
```

More complete examples

The following examples demonstrate a range of mathematical content typeset using LaTeX.

Subscripts in math mode are written as a_b and superscripts are written as a^b. These can be combined and nested to write expressions such as

$$T^{i_1 i_2 \ldots i_p}_{j_1 j_2 \ldots j_q} = T(x^{i_1}, \ldots, x^{i_p}, e_{j_1}, \ldots, e_{j_q})$$

We write integrals using $\int$ and fractions using $\frac{a}{b}$. Limits are placed on integrals using superscripts and subscripts:

$$\int_0^1 \frac{dx}{e^x} = \frac{e-1}{e}$$

Lower case Greek letters are written as ω δ etc. while upper case Greek letters are written as Ω Δ.
Mathematical operators are prefixed with a backslash as $\sin(\beta)$, $\cos(\alpha)$, $\log(x)$ etc.

amsmath equation* environment

The next example uses the equation* environment which is provided by the amsmath package, so we need to add the following line to our document preamble:

```
\usepackage{amsmath}% For the equation* environment
```

amsmath equation* environment

```
\documentclass[12pt, a4paper]{article}
\usepackage{amsmath}% For the equation* environment
\begin{document}
\section{First example}
     The well-known Pythagorean theorem \(x^2 + y^2 = z^2\) was proved to be
invalid for other exponents, meaning the next equation has no integer solutions for
\(n>2\):
     \[ x^n + y^n = z^n \]
\section{Second example}
     This is a simple math expression \(\sqrt{x^2+1}\) inside text.
     And this is also the same:
     \begin{math}
          \sqrt{x^2+1}
     \end{math}
     but by using another command.
     This is a simple math expression without numbering
     \[\sqrt{x^2 + 1}n]
     separated from text.
     This is also the same:
     \begin{displaymath}
          \sqrt{x^2+1}
     \end{displaymath}
     \ldots and this:
     \begin{equation*}
          \sqrt{x^2 + 1}
     \end{equation*}
\end{document
```

More complete examples

First example

The well-known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents, meaning the next equation has no integer solutions for $n > 2$:

$$x^n + y^n = z^n$$

Second example

This is a simple math expression $\sqrt{x^2 + 1}$ inside text.
And this is also the same:
$\sqrt{x^2 + 1}$ but by using another command.
This is a simple math expression without numbering

$$\sqrt{x^2 + 1}$$

separated from text.
This is also the same:

$$\sqrt{x^2 + 1}$$

. . . and this:

$$\sqrt{x^2 + 1}$$

Reference guide

Below is a table with some common maths symbols. For a more complete list see the List of Greek letters and math symbols:

description	code	examples
Greek letters	`\alpha \beta \gamma \rho \sigma \delta \epsilon`	$\alpha \beta \gamma \rho \sigma \delta \epsilon$
Binary operators	`\times \otimes \oplus \cup \cap`	$\times \otimes \oplus \cup \cap$
Relation operators	`< > \subset \supset \subseteq \supseteq`	$< > \subset \supset \subseteq \supseteq$
Others	`\int \oint \sum \prod`	$\int \oint \sum \prod$

Different classes of mathematical symbols are characterized by different formatting (for example, variables are italicized, but operators are not) and different spacing.

Greek letters

αA	`\alpha A`	νN	`\nu N`
βB	`\beta B`	$\xi \Xi$	`\xi \Xi`
$\gamma \Gamma$	`\gamma \Gamma`	oO	`o O`
$\delta \Delta$	`\delta \Delta`	$\pi \Pi$	`\pi \Pi`
$\epsilon \varepsilon E$	`\epsilon \varepsilon E`	$\rho \varrho P$	`\rho \varrho P`
ζZ	`\zeta Z`	$\sigma \Sigma$	`\sigma \Sigma`
ηH	`\eta H`	τT	`\tau T`
$\theta \vartheta \Theta$	`\theta \vartheta \Theta`	$\upsilon \Upsilon$	`\upsilon \Upsilon`
ιI	`\iota I`	$\phi \varphi \Phi$	`\phi \varphi \Phi`
κK	`\kappa K`	χX	`\chi X`
$\lambda \Lambda$	`\lambda \Lambda`	$\psi \Psi$	`\psi \Psi`
μM	`\mu M`	$\omega \Omega$	`\omega \Omega`

Arrows

$\leftarrow$	\leftarrow	$\Leftarrow$	\Leftarrow
$\rightarrow$	\rightarrow	$\Rightarrow$	\Rightarrow
$\leftrightarrow$	\leftrightarrow	$\rightleftharpoons$	\rightleftharpoons
$\uparrow$	\uparrow	$\downarrow$	\downarrow
$\Uparrow$	\Uparrow	$\Downarrow$	\Downarrow
$\Leftrightarrow$	\Leftrightarrow	$\Updownarrow$	\Updownarrow
$\mapsto$	\mapsto	$\longmapsto$	\longmapsto
$\nearrow$	\nearrow	$\searrow$	\searrow
$\swarrow$	\swarrow	$\nwarrow$	\nwarrow
$\leftharpoonup$	\leftharpoonup	$\rightharpoonup$	\rightharpoonup
$\leftharpoondown$	\leftharpoondown	$\rightharpoondown$	\rightharpoondown

Miscellaneous symbols

∞	\infty	$\forall$	\forall
$\Re$	\Re	$\Im$	\Im
∇	\nabla	$\exists$	\exists
∂	\partial	$\nexists$	\nexists
$\emptyset$	\emptyset	$\varnothing$	\varnothing
$\wp$	\wp	$\complement$	\complement
$\neg$	\neg	$\cdots$	\cdots
$\square$	\square	$\surd$	\surd
$\blacksquare$	\blacksquare	$\triangle$	\triangle

Binary Operation/Relation Symbols

$\times$	\times	$\cdot$	\cdot
$\div$	\div	$\cap$	\cap
$\cup$	\cup	$\neq$	\neq
$\leq$	\leq	$\geq$	\geq
$\in$	\in	$\perp$	\perp
$\notin$	\notin	$\subset$	\subset
$\simeq$	\simeq	$\approx$	\approx
$\wedge$	\wedge	$\vee$	\vee
$\oplus$	\oplus	$\otimes$	\otimes
$\Box$	\Box	$\boxtimes$	\boxtimes
$\equiv$	\equiv	$\cong$	\cong

Brackets and Parentheses

Parentheses and brackets are very common in mathematical formulas. You can easily control the size and style of brackets in LaTeX; this article explains how. Here's an table of listing some common math braces and parentheses used in LaTeX:

$\times$	\times	$\cdot$	\cdot
$\div$	\div	$\cap$	\cap
$\cup$	\cup	$\neq$	\neq
$\leq$	\leq	$\geq$	\geq
$\in$	\in	$\perp$	\perp
$\notin$	\notin	$\subset$	\subset
$\simeq$	\simeq	$\approx$	\approx
$\wedge$	\wedge	$\vee$	\vee
$\oplus$	\oplus	$\otimes$	\otimes
$\Box$	\Box	$\boxtimes$	\boxtimes

Some examples

The size of brackets and parentheses can be manually set, or they can be resized dynamically in your document, as shown in the next example:

```
\[
F = G \left( \frac{m_1 m_2}{r^2} \right)
\]
```

$$F = G \left(\frac{m_1 m_2}{r^2} \right)$$

Some examples

$\dots$	$\left[\dfrac{N}{\left(\frac{L}{p}\right) - (m+n)} \right]$

Some examples

When writing multi-line equations with the align, align* or aligned environments, the \left and \right commands must be balanced on each line and on the same side of &. Therefore the following code snippet will fail with errors:

```
\begin{align*}
y  = 1 + & \left(  \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \ldots \right. \\
  &\left. \quad + \frac{1}{x^{n-1}} + \frac{1}{x^n} \right)
\end{align*}
```

$$y = 1 + \left(\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \ldots + \frac{1}{x^{n-1}} + \frac{1}{x^n} \right)$$

Controlling types and sizes

`\[ \Biggl \langle 3x+7 \biggr \rangle \]`	$\Biggl \langle 3x+7 \biggr \rangle$

Reference guide

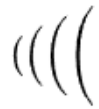
LaTeX markup	Renders as
`\bigl( \Bigl( \biggl( \Biggl(`	
`\bigr] \Bigr] \biggr] \Biggr]`	
`\bigl\{ \Bigl\{ \biggl\{ \Biggl\{`	
`\bigl \langle \Bigl \langle \biggl \langle \Biggl \langle`	
`\bigr \rangle \Bigr \rangle \biggr \rangle \Biggr \rangle`	
`\big| \Big| \bigg| \Bigg|`	
`\big\vert \Big\vert \bigg\vert \Bigg\vert`	
`\bigl \lceil \Bigl \lceil \biggl \lceil \Biggl \lceil`	
`\bigr \rceil \Bigr \rceil \biggr \rceil \Biggr \rceil`	

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Thank you!

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